Mohamed Omrani

Embedded Software Engineer — STM32, RTOS & Edge AI

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Tunis, Tunisia · Available immediately

Profile

Embedded software engineer working where hardware and software meet: firmware close to the metal, and the data processing that turns raw sensor streams into a decision someone can act on. Most at home on systems that have to be correct — an actuator that must stop, a measurement that must be trusted, a fault caught before it becomes a breakdown. Now looking for an embedded role on a physical product.

Work Experience

Predictive Maintenance Platform for Vehicles — AutoTwin *Capgemini Engineering, Tunisia — Feb 2026 – Sep 2026*

- OBD-II acquisition front-end on **STM32F746G-DISCO** (FreeRTOS, TouchGFX, DMA UART, USB-HID): 20 live vehicle channels at 10 Hz, per-sensor fault injection, in-dash alarm overlay, and a 16-byte AI-verdict channel back to the board.
- Diagnostic engine on Raspberry Pi: **20 XGBoost "digital twins"**, each sensor predicted from all the others; the residuals feed a self-calibrating baseline that isolates the root cause and forecasts remaining useful life in kilometres.
- Results on a held-out vehicle: **9/10 faults caught, 0 false positives, 40/46 vs 29/46** against the static baseline. Also delivered the marketplace stack (Spring Boot 3, Angular 18, MongoDB, Firebase, two Flutter apps) and the test qualification suite (40/40 passed).

Embedded Systems Intern *COFAT Group, Tunis — Jul–Aug 2025*

- Designed a dynamic stress-testing machine in SolidWorks (ISO functional dimensioning) and wrote its **FreeRTOS/STM32** firmware: stepper control, OLED UI, emergency-stop safety.
- Low-level peripheral drivers and hardware-near control loops, validated with lab instrumentation (oscilloscope, logic analyzer).

Web Development Intern *Wassali Express, Tunis — Jul–Aug 2024*

- Transport Management System: Spring Boot REST APIs, Angular UI, MongoDB persistence, automated API testing with Postman.

Education

2022 – Sep 2026 **National Engineering Diploma — Embedded Systems and IoT**, ISIMG, University of Gabès
2021 **Baccalaureate in Mathematics (German option)**, Lycée Jemmal

Technical Skills

Embedded	STM32F7/F1, HAL and register-level drivers, FreeRTOS, TouchGFX, DMA, UART/SPI/I2C/CAN/OBD-II, JTAG, low-power and memory-constrained design
Edge AI	PyTorch, YOLO, scikit-learn, XGBoost, OpenCV, feature engineering, on-device inference (Raspberry Pi, MCU)
Validation	GCC + Makefile, GDB/ST-Link, logic analyzer and oscilloscope, unit and integration tests, Git, CI, test design
Backend	Python (FastAPI), Java 17, Spring Boot 3, Angular, Flutter, REST, MongoDB, Firebase, SQL
Other	VHDL/FPGA, BLE/LoRa/MQTT, AES-128/HMAC-SHA256, SolidWorks

Open-Source Projects

- **cable-endurance-bench** — Cable endurance test bench on STM32F103: cycle parameters, OLED status, stepper drive, E-stop and guard interlock (embedded C, HAL).
- **secure-field-monitoring** — LoRa sensor link sealed with AES-128 + HMAC-SHA256, ESP32 to Raspberry Pi.
- **live-object-detection** — Real-time YOLO detection on a Raspberry Pi, exported to NCNN (Python).
- **access-code-lock** — Combination lock in VHDL on an Intel MAX 10: FSM, debouncing, 7-segment.

Certifications

NVIDIA Deep Learning Fundamentals (2024) · Evaluation & Customization of LLMs (2025) · Generative AI with Diffusion Models (2024) · SCRUMstudy Scrum Fundamentals (2024)

Languages

Arabic (native) · French (fluent) · English (professional) · German (A2)